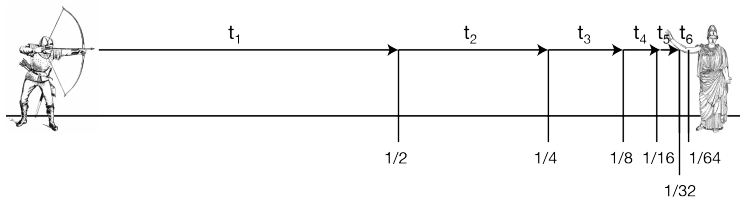


Topic 2 - Real series

Calculus



- Let the distance s be 1 and assume that we have a constant velocity $v := 1$.
- Total time needed to make the distance: $T = \frac{s}{v} = 1$, which is a finite number.
- Note that the movement of the arrow can be divided into an infinite number of steps:
 - ★ We need $t_1 = \frac{s_1}{v} = \frac{1}{2}$ to cover half the distance.
 - ★ Then we need $t_2 = \frac{1}{4}$ to cover the next part, which is the quarter of the total distance.
 - ★ Then we need $t_3 = \frac{1}{8}$ to cover the next part.
 - ⋮
 - ★ To take the n -th step: we need a time $t_n = \frac{1}{2^n}$.
 - ⋮
- *Question:* Is the equality $t_1 + t_2 + t_3 + \cdots + t_n + \cdots = T = 1$ true?

- So in fact we would like to interpret the sum of the members of a real sequence (t_n) :
 - ★ How to interpret a sum containing infinite (countable) number of members?
 - ★ Having interpreted this, how do we calculate its value?
 - ★ *It would not prove the validity of the interpretation, but it would provide some support for it, if in the above example we actually received 1 for the value.*
 - ★ Does a sum with infinitely many members have the usual properties of sums with finitely many members (commutativity and associativity, factorization)?

- *Real series:*

For a sequence $(a_n) \subseteq \mathbb{R}$, $\longrightarrow$ the question is: $a_1 + a_2 + a_3 + \cdots + a_n + \cdots = ?$

- *Idea:* Since we can already treat finite sums, let's approximate the above value by always summing only finitely many members:

- ★ the value $s_1 := a_1$ will still be far from the true result,
- ★ $s_{10} := a_1 + \cdots + a_{10}$ should already be a "better", in fact,
- ★ $s_{10^{100}} := a_1 + \cdots + a_{10^{100}}$ is "much better" ... etc.

Real series

- Let $(a_n) \subseteq \mathbb{R}$ be a real sequence and

$$\sum_{k=1}^1 a_k := a_1, \quad \text{and} \quad \sum_{k=1}^n a_k := a_n + \sum_{k=1}^{n-1} a_k \quad \text{if } n \geq 2.$$

- Definition (Real Series).** Keeping the above notations, the real sequence

$$\left(\sum_{k=1}^n a_k \right) \subseteq \mathbb{R} \tag{1}$$

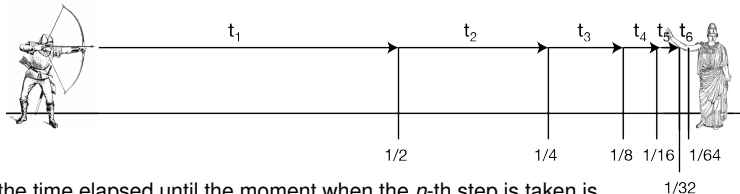
is called *a real series defined by (a_n)* , furthermore the n th member of the above sequence is the *n th partial sum* of the series in question.

For simplicity, For the real number sequence (1), we often refer as $\sum a_n$.

- If (1) is convergent, then its limit is called the *sum* of $\sum a_n$ and denoted by

$$\sum_{n=1}^{\infty} a_n.$$

Geometric series



So the time elapsed until the moment when the n -th step is taken is

$$s_n := t_1 + t_2 + t_3 + \cdots + t_n = \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \cdots + \frac{1}{2^n} = \sum_{k=1}^n \frac{1}{2^k}.$$

$$s_n = \sum_{k=1}^n \frac{1}{2^k} = \frac{1}{2^n} \sum_{k=1}^n 2^{k-1} = \frac{2^n - 1}{2^n} = 1 - \frac{1}{2^n} \rightarrow 1 \text{ if } n \rightarrow \infty.$$

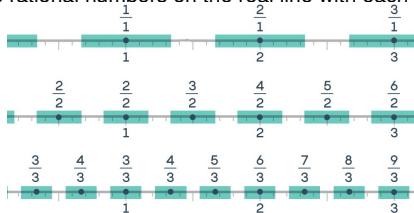
So there is no contradiction, the time until the arrow strikes is $T = 1$.

• **Theorem (Geometric series).** Let $q \in \mathbb{R}$. The real series defined by the sequence (q^n) is called an geometric series, which is convergent if and only if $|q| < 1$ holds, and, in this case, we have

$$\sum_{n=1}^{\infty} q^n = \frac{q}{1-q}.$$

Geometric series

- *Problem:* Suppose you have a 1 cm length of duct tape. Can we cut it up so that we can glue/cover all the rational numbers on the real line with each piece?



- *Precisely:* Can the set of rational numbers be covered by a family of intervals whose members are all of positive length and whose total length is 1?
- *More precisely:* Is there $\mathcal{J} := \{I \subseteq \mathbb{R} \mid I \text{ is an interval of positive length}\}$ such that

$$\mathbb{Q} \subseteq \bigcup_{I \in \mathcal{J}} I \quad \text{and} \quad \sum_{I \in \mathcal{J}} \text{diam}(I) = 1?$$

Harmonic series

The real series defined by the sequence $(\frac{1}{n})$ is called a harmonic series.

• **Proposition.** The harmonic series has no sum.

Indirectly, assume that it exists: $S > 0$. Let $a_n := \frac{1}{n}$ if $n \in \mathbb{N}$ and define (a_n^*) as

$$a_1^* := a_1, \quad a_2^* := a_2, \quad \text{and} \quad a_n^* := \begin{cases} a_{n+1} & \text{if } n \geq 3 \text{ odd,} \\ a_n & \text{if } n \geq 3 \text{ even.} \end{cases}$$

Then $a_n^* \leq a_n$ for all $n \in \mathbb{N}$ and the inequality is strict whenever $n \geq 3$ is odd. Thus

$$s_n^* := \sum_{k=1}^n a_k^* < \sum_{k=1}^n a_k =: s_n, \quad (n \in \mathbb{N} \text{ and } n \geq 3).$$

It is easy to see that $s_{2n}^* = s_n + \frac{1}{2}$ for all $n \in \mathbb{N}$, consequently

$$S = \lim_{n \rightarrow \infty} s_n = \lim_{n \rightarrow \infty} s_{2n} \geq \lim_{n \rightarrow \infty} s_{2n}^* = \lim_{n \rightarrow \infty} (s_n + \frac{1}{2}) = S + \frac{1}{2}.$$

Since S is real, it follows that $\frac{1}{2} \leq 0$, which is a contradiction.

Harmonic series

So the harmonic series is divergent, or we can say that the sum exists only in an extended sense:

$$\sum_{n=1}^{\infty} \frac{1}{n} = +\infty.$$

- *Illustratively:* If you were to build a walkway between the Moon and Earth, and you started out on foot by first walking 1 metres, then only half a metre, then $\frac{1}{3}$ metres, and so on, you would eventually, after some number of steps, reach the surface of the Moon. In fact, we could even build a walkway between the Earth and the boundary of the visible Universe!

- *Basel Problem (no problem today!):* The series constructed from the reciprocals of the squares of natural numbers is not divergent!

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}.$$

Necessary condition for convergence

- **Theorem (Necessary condition for convergence).** If $\sum a_n$ is convergent, then (a_n) is a null sequence.
- The behavior of the harmonic series shows that the converse of the theorem is not true, i.e., the fact that the general term tends to zero does not imply convergence of the series.
- Thus the above theorem can only be used to prove divergence: if the general member is not a null sequence, then the series cannot be convergent.
- *Show that $\sum (-1)^n$ has no sum!*
 - ★ *Using the definition.* The sequence (s_n) has two different accumulation points, so the series cannot be convergent. Indeed: for all $n \in \mathbb{N}$, $s_{2n} = 0$ and $s_{2n-1} = -1$.
 - ★ *Using the necessary condition.* The general member, i.e. the $(-1)^n$ does not tend to zero as $n \rightarrow \infty$, so the series cannot be convergent.

So this series is not divergent like the harmonic series: the sum here is not meaningful at all.

Absolutely and conditionally convergent series

- **Definition (Absolutely convergent series).** A series $\sum a_n$ is called *absolutely convergent* if $\sum |a_n|$ is convergent.
- **Theorem (Connection of convergent and absolutely convergent series).** If a real sequence is absolutely convergent, then it is convergent.
- This is because the sequence of partial sums of an absolutely convergent series is a Cauchy sequence and therefore convergent.
- The converse of the theorem is not true: there is a convergent real series that is not absolutely convergent: *alternating harmonic series*;

$$\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} = \ln 2.$$

- **Definition (Conditionally convergent series).** If a series is convergent but it is not absolutely convergent then it is called *conditionally convergent*.

„Commutativity” - Rearrangement of a series

Let $U, V \subseteq \mathbb{R}$ be nonempty. We shall say that a function $f : D \rightarrow V$ is *surjective* if, for all $v \in V$, there exists $u \in U$ such that $f(u) = v$. The function f is called *injective* if, for all $u_1, u_2 \in U$ with $u_1 \neq u_2$ we have $f(u_1) \neq f(u_2)$. Finally f is *bijective* if it is surjective and injective.

• **Definition (Rearrangement of a series).** Let $\varphi : \mathbb{N} \rightarrow \mathbb{N}$ be a bijective function. Then the series $\sum a_{\varphi(n)}$ is a rearrangement of $\sum a_n$.

• **Theorem (Rearrangement of an absolutely convergent series).** If a series is absolutely convergent then all of its rearrangement series is convergent.

• **Tétel (Riemann's Rearrangement Theorem).** If $\sum a_n$ is conditionally convergent, then, for all $-\infty \leq \alpha \leq \beta \leq +\infty$, there exists a bijection $\varphi : \mathbb{N} \rightarrow \mathbb{N}$ such that

$$\liminf_{n \rightarrow \infty} \sum_{k=1}^n a_{\varphi(n)} = \alpha \quad \text{and} \quad \limsup_{n \rightarrow \infty} \sum_{k=1}^n a_{\varphi(n)} = \beta.$$

„Associativity” - Grouping of a series

- **Definition (Grouping of a series).** Let $(a_n) \subseteq \mathbb{R}$ be a real sequence and $(x_n) \subseteq \mathbb{N}$ be strictly increasing, furthermore let $x_0 := 0$. Then the series

$$\sum_{n=1}^{\infty} \left(\sum_{k=x_{n-1}+1}^{x_n} a_k \right)$$

is called a grouped series of $\sum a_n$.

Tétel (Grouping of a convergent series). Any grouped series of a convergent series is convergent having the same sum of the initial series.

- There are divergent real series for which a grouped series is convergent:

$$\text{For } \sum_{n=1}^{\infty} (-1)^n \text{ let } x_n := 2n, \text{ if } n \in \mathbb{N},$$

$$\text{then: } \sum_{k=x_{n-1}+1}^{x_n} (-1)^k = (-1)^{2n-1} + (-1)^{2n} = 0.$$

Operations and convergence

• **Theorem (Operations and convergence).** Let $\sum a_n$ and $\sum b_n$ be convergent with a sum $A \in \mathbb{R}$ and $B \in \mathbb{R}$, respectively, and let $\lambda \in \mathbb{R}$. Then the following statements are true.

(1) The sum $\sum(a_n + b_n)$ of $\sum a_n$ and $\sum b_n$ is convergent and its sum is $A + B$.

(2) The scalar multiple $\sum(\lambda a_n)$ of $\sum a_n$ is convergent and its sum is λA .

Finally, Minkowski's Inequality holds, that is, we have

$$\left| \sum_{n=1}^{\infty} a_n \right| \leq \sum_{n=1}^{\infty} |a_n|.$$

• **Definition (Cauchy-product of two real series).** The Cauchy product of the real series $\sum a_n$ and $\sum b_n$ is $\sum c_n$, where

$$c_n := \sum_{k=1}^n a_k b_{n+1-k}, \quad n \in \mathbb{N}.$$

• **Theorem (Mertens Theorem).** If two real series are convergent and at least one of them is absolutely convergent, then their Cauchy product is convergent.

Convergence criteria

- The following theorems can decide for a set of real series whether they are convergent or not.
- In the convergent case, however, they say nothing about the value of the sum!
- If you also want to know the value of the sum, you can use the definition (limit of the sequence of partial sums), but this is not always easy.
- In general, your approach can be the following:
 - ★ *Check that the necessary condition for convergence is satisfied.* If not, then the series cannot be convergent: we are done. If it is: we can't say anything for sure about convergence (see exponential series and harmonic series), so further calculation is needed.
 - ★ Let's apply the appropriate convergence criterion (the shape of the general term helps a lot)!

(1) Comparison Test

• **Theorem (Comparison Test).** Let $(a_n), (b_n) \subseteq \mathbb{R}$ be sequences with non-negative members and assume that we have $a_n \leq b_n$ for all $n \in \mathbb{N}$. Then the following statements hold.

1. If $\sum b_n$ has a sum, then $\sum a_n$ has a sum.
2. If $\sum a_n$ has no sum, then $\sum b_n$ has no sum.

• To apply the comparison test, we must first guess whether our series is convergent or divergent.

• If we think of it as convergent, we have to use the 1. statement. If divergent, then statement 2. must be applied.

• *Investigate $\sum \frac{2n^2+3n+1}{4n^4+n^3+10}$ for convergence!*

For a large n , the numerator and denominator will be close to $2n^2$ and $4n^4$ respectively. Hence, for large n , we almost have $\frac{2n^2}{4n^4} = \frac{1}{2} \cdot \frac{1}{n^2}$. The series $(\frac{1}{n^2})$ defines a convergent series, so the conjecture is: our original series is convergent.

$$a_n := \frac{2n^2 + 3n + 1}{4n^4 + n^3 + 10} \leq \frac{2n^2 + 3n^2 + n^2}{4n^4 + n^3 + 10} \leq \frac{6n^2}{4n^4} = \frac{3}{2} \cdot \frac{1}{n^2} =: b_n.$$

Thus the original series is convergent, moreover, for the sum S , we obtained that

$$0 < S \leq \frac{\pi^2}{4} \approx \frac{9}{4}.$$

(2) Condensation Test

• **Theorem (Condensation Test).** Let $(a_n) \subseteq \mathbb{R}$ be a real sequence with non-negative members and assume that it is decreasing. The series $\sum a_n$ has a sum if and only if $\sum 2^n a_{2^n}$ has a sum, moreover in case of convergence the sum $\sum 2^n a_{2^n}$ is not larger than the double of the sum of the original series.

• So here we replace the original series with a new series, which we hope will be easier to examine. The point is that we do this in such a way that the new series behaves, from a convergence point of view, in the same way as the original!

• The harmonic series satisfies the conditions of the above theorem, so it behaves in the same way as

$$\sum_{n=1}^{\infty} 2^n \cdot \frac{1}{2^n} = \sum_{n=1}^{\infty} 1,$$

which is obviously divergent, because its general term is not a null sequence.

• Investigate $\sum_{n=2}^{\infty} \frac{1}{n \log_3(n)}$ for convergence!

The conditions are met, thus consider

$$\sum_{n=2}^{\infty} 2^n \cdot \frac{1}{2^n \log_3(2^n)} = \sum_{n=2}^{\infty} \frac{1}{\log_3(2^n)} = \sum_{n=2}^{\infty} \frac{1}{n \log_3 2} = \frac{1}{\log_3 2} \sum_{n=2}^{\infty} \frac{1}{n}.$$

Our original series is therefore divergent. *(In the last step, we took advantage of the fact that the common constant multiplier of the members can be factorized - regardless of the existence of the sum.)*

(3) Leibniz's Test

• **Theorem (Leibniz's Test).** If (a_n) is a decreasing null sequence, then $\sum (-1)^n a_n$ is convergent.

• Apply Leibniz's Test for the alternating harmonic series!

• Investigate $\sum (-1)^n (\sqrt{n+1} - \sqrt{n})$ for convergence!

The series is alternating, so in order to apply Leibniz's criterion, we need to consider the series $(a_n := \sqrt{n+1} - \sqrt{n})$ in terms of monotonicity and convergence.

★ Monotonicity:

$$a_{n+1} \leq a_n \quad \Leftrightarrow \quad 0 \leq 1.$$

★ Convergence:

$$a_n = \frac{1}{\sqrt{n+1} + \sqrt{n}} \rightarrow 0, \quad \text{if } n \rightarrow \infty.$$

• *Cross-examination:* Is $\sum (\sqrt{n+1} - \sqrt{n})$ convergent?

(4) Root Test

- **Theorem (Cauchy's Root Test).** For a given real sequence (a_n) , let

$$C := \limsup_{n \rightarrow \infty} \sqrt[n]{|a_n|} \in \mathbb{R} \cup \{-\infty, +\infty\}.$$

1. If $C < 1$, then $\sum a_n$ is absolutely convergent.
 2. If $C > 1$, then $\sum a_n$ is divergent.
- From what we learned earlier, C is well-defined for any real sequence of numbers.
 - *Message of the above theorem:* Using the general term of the series, we construct a sequence whose convergence we examine. This is what we would do if we were examining series convergence by definition, but the resulting auxiliary series is usually easier to examine.
 - *In general:* root test is effective if the general term contains a n -th power.

(5) Ratio Test

- **Theorem (D'Alembert's Ratio Test).** Let (a_n) be a real sequence with non-zero terms and let

$$D := \limsup_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| \in \mathbb{R} \cup \{-\infty, +\infty\}.$$

1. If $D < 1$, then $\sum a_n$ is absolutely convergent.
 2. If $D > 1$, then $\sum a_n$ is divergent.
- From what we learned earlier, D is well-defined for any real sequence of numbers.
 - *Message of the above theorem:* Using the general term of the series, we construct a sequence whose convergence we examine. This is what we would do if we were examining series convergence by definition, but the resulting auxiliary series is usually easier to examine.
 - *In general:* ratio test is effective if the general term contains factorial or n -th power.